

Algebraic and Transcendental functions

Professor: Francisco Vazquez-Tavares

Last compilation: November 11, 2025

Name:

Algebra skills

Combine the functions $f(x) = x^2 + 1$ and $g(x) = x - 4$, according to the indicated operation,

- $(f + g)(x)$
- $\left(\frac{f}{g}\right)(0)$
Solve the following equations,
- $\log_2(2x + 2) - \log_2(x + 1) = 3$
- $2\log_3(x) - \log_3(x - 4) = 2 + \log_3(2)$

Function analysis

For each of the following pair of function, find the indicated composition, its domain and range.

- $f(x) = x^2 + 7$ and $g(x) = \sqrt{x + 5}$
- $f(x) = \frac{7x}{x+2}$ and $g(x) = \frac{5}{x}$

Find all the asymptotes, x and y interceptions of the following function,

$$f(x) = \frac{2x - 4}{2x^2 + x - 10}$$

Modelling with math

Publishing company. A small publishing company is releasing a new book. The production costs will include a one-time fixed cost for editing and an additional cost for each book printed. The total production cost C (in dollars) is given by the function $C = 650 + 15.9N$, where N is the number of books. The total revenue earned (in dollars) from selling the books is given by the function $R = 32.80N$. Let P be the profit made (in dollars).

- Write an equation relating P to N .
- Simplify your answer as much as possible.

Population growth. The population of a certain city is given by $P(t) = 234e^{0.071t}$, where P is measured in thousands of people and t is measured in years from the year 2000.

- Find the population for the year 2014.
- Find the population for the year 2027. find when the population will be 3 million.

Algebra Toolkit

Algebra properties

$$a + b = b + a,$$

$$(a + b) + c = a + (b + c),$$

$$a(b + c) = ab + ac,$$

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}, \quad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd},$$

$$\frac{-a}{b} = -\frac{a}{b} = \frac{a}{-b},$$

$$\frac{\frac{a}{b}}{\frac{c}{d}} = \frac{ad}{bc}$$

$$\sqrt{ab} = \sqrt{a}\sqrt{b}, \quad \sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}, \quad \sqrt[n]{a} = a^{1/n}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a},$$

$$a^2 - b^2 = (a - b)(a + b)$$

Laws of exponents

$$a^r \cdot a^s = a^{r+s}, \quad (a^r)^s = a^{rs}$$

$$\frac{a^r}{a^s} = a^{r-s}, \quad (ab)^r = a^r b^r$$

$$\left(\frac{a}{b}\right)^r = \frac{a^r}{b^r}, \quad b \neq 0$$

Laws of Logarithms

$$\log_a(AB) = \log_a(A) + \log_a(B)$$

$$\log_a\left(\frac{A}{B}\right) = \log_a(A) - \log_a(B)$$

$$\log_a(A^C) = C \log_a(A)$$

Inverse functions property

$$(f \circ f^{-1})(x) = x = (f^{-1} \circ f)(x)$$

$$\log_a(a^x) = x = a^{\log_a(x)}$$

Conic sections

Identify the type of conic section and find the following,

- If is a circle,
 1. Center
 2. Radius
- If is parabola
 1. Vertex
 2. Focus
 3. Directrix
- If is an ellipse
 1. Center
 2. Focus (2)
 3. Semi-axis (2)
- If is a hyperbola
 1. Center
 2. Focus (2)
 3. Asymptotes (2)

1. $x^2 - 4x + 4 + y^2 + 6y + 9 = 9$
2. $x^2 + (y - 2)^2 = 36$

Conic Sections

Standard form of a circle

$$(x - h)^2 + (y - k)^2 = r^2$$

Standard form of a parabola

$$(x - h) = 4a(y - k)^2, \text{ or } (y - k) = 4a(x - h)^2$$

Standard form of an ellipse

$$\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1$$

Standard form of a hyperbola

$$\frac{(x - h)^2}{a^2} - \frac{(y - k)^2}{b^2} = 1$$